Air University
Final Examinations: Spring 2026

Student ID: [redacted]

Subject: Applied Physics
Class: BSCYS, BSIT, BSCS-additional students
Section(s): II-A
Course Code: PH-111

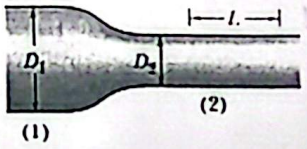
Time Allowed: 03 hours
Max Marks: 100
FM's Name: Dr. Saima Rafique
FM's Signature:

INSTRUCTIONS

- Attempt responses in the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on the question paper will be considered as the use of unfair means.
- Calculators are allowed.

Q1. Marks (15+10): (PLO-2: CLO-2)

(a) Figure shows wire section 1 of diameter $D_1 = 4.00R$ and wire section 2 of diameter $D_2 = 2.00R$, connected by a tapered section. The wire is copper and carries a current. Assume that the current is uniformly distributed across any cross-sectional area through the wire's width. The electric potential change V along the length L =2.00 m shown in section 2 is 10.0 μV. The number of charge carriers per unit volume is $8.49 \times 10^{28}\ m^{-3}$. What is the drift speed of the conduction electrons in section 1?



(b) The current density in a wire is uniform and has magnitude 2.0 106 A/m^2, the wire's length is 5 m, and the density of conduction electrons is $8.49 \times 10^{28}\ m^{-3}$. How long does an electron take (on average) to travel the length of the wire?

Q2. Marks (10+10): (PLO-1: CLO-1)

(a) A beam of electrons is projected into a chamber by an *electron gun* G. The electrons enter in the plane of the page with speed v and then move in a region of uniform magnetic field directed out of that plane. As a result, a magnetic force continuously deflects the electrons, and are always perpendicular to each other, this deflection causes the electrons to follow a circular path. For this circular motion evaluate radius (r), the period (T), frequency (f), and angular frequency (ω).

(b) An electron with a kinetic energy of 22.5 eV moves into a region of uniform magnetic field (B) of magnitude 4.55×10^{-4} T. The angle between the directions of and the electron's velocity is 65.5°. What is the pitch of the helical path taken by the electron?

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Q3. Marks (10+15): (PLO-2: CLO-2)

(a) Consider an infinitely long straight wire carrying a constant current I. Using the elemental contribution dB from the Biot–Savart law, obtain the total magnetic field at a point perpendicular distance R from the wire. Draw the required diagram and explain each mathematical step carefully.

(b) Consider a circular wire loop of radius a located in the yz plane and carrying a steady current I as in Figure 1. Calculate the magnetic field at an axial point P a distance x from the centre of the loop (using Biot-savart law).

Q4. Marks (10+10): (PLO-3: CLO-3)

(a) Figure 2 shows the cross section of a long conducting cylinder with inner radius $a = 2$ cm and outer radius $b = 4$ cm. The cylinder carries a current out of the page, and the magnitude of the current density in the cross section is given by $J = cr^2$, with $c = 3.0 \times 10^6$ A/m^4 and r in meters. What is the magnetic field at the dot in Fig. 2, which is at radius $r = 3$ cm from the central axis of the cylinder?

(b) A cylindrical conductor with radius R carries a current I (Fig. 3). The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnetic field as a function of the distance r using Amperes law from the conductor axis for points both inside and outside the

Figure 1

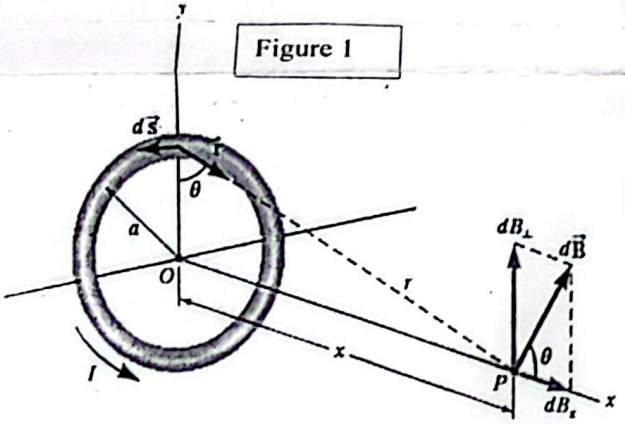


Figure 2

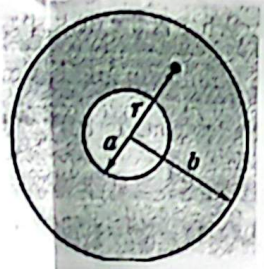
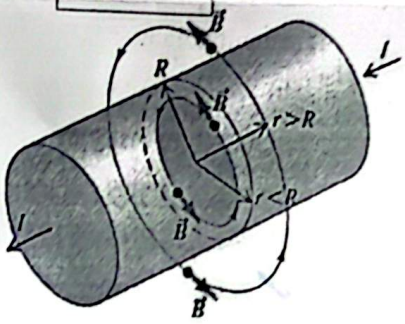
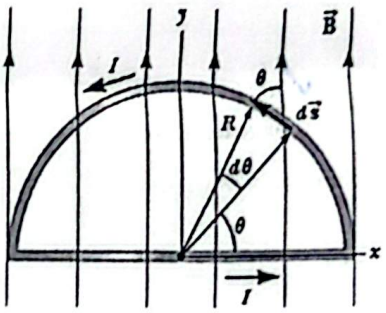


Figure 3



Q5. Marks (10): (PLO-3: CLO-3)

A wire bent into a semicircle of radius R forms a closed circuit and carries a current I. The wire lies in the xy plane, and a uniform magnetic field is directed along the positive y axis as in Figure. Find the magnitude and direction of the magnetic force acting on the straight portion of the wire and on the curved portion.



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